Multivariable Calculus
Problem Set 8: Solutions

Angad Singh Ahuja

April 19, 2026

Note. The source PDF is *Problem Set 8*. I have placed the solutions in the folder name requested by the prompt, namely `PS_6_Solutions`. Also, in Problem 3(i) the PDF appears to have a missing minus sign; the region is interpreted as $-2 \le x \le 2$, $1 \le y \le 6$.

Problem 1

Find the extreme values of f subject to both constraints.

(i) $f(x, y, z) = x + 2y$ subject to $x + y + z = 1$ and $y^2 + z^2 = 4$.

Solution. From the linear constraint,

$$x = 1 - y - z.$$

Hence

$$f = x + 2y = (1 - y - z) + 2y = 1 + y - z.$$

So we need to maximize/minimize $y - z$ on the circle $y^2 + z^2 = 4$. Write

$$y - z = \langle (y, z), (1, -1) \rangle.$$

By Cauchy-Schwarz,

$$|y - z| \le \sqrt{y^2 + z^2}\,\sqrt{1^2 + (-1)^2} = 2\sqrt{2}.$$

Therefore

$$f_{\max} = 1 + 2\sqrt{2}, \qquad f_{\min} = 1 - 2\sqrt{2}.$$

Equality holds when (y, z) is parallel to $(1, -1)$ or $(-1, 1)$:

$$(y, z) = (\sqrt{2}, -\sqrt{2}) \quad \text{or} \quad (-\sqrt{2}, \sqrt{2}).$$

Since $x = 1 - y - z$, in both cases $x = 1$. Hence

$$f_{\max} = 1 + 2\sqrt{2} \text{ at } (1, \sqrt{2}, -\sqrt{2}),$$

$$f_{\min} = 1 - 2\sqrt{2} \text{ at } (1, -\sqrt{2}, \sqrt{2}).$$

(ii) $f(x,y,z) = 3x - y - 3z$ subject to $x + y - z = 0$ and $x^2 + 2z^2 = 1$.

Solution. From $x + y - z = 0$ we get

$$y = z - x.$$

Then

$$f = 3x - (z - x) - 3z = 4x - 4z = 4(x - z).$$

Thus we maximize/minimize the linear function $x - z$ on the ellipse

$$x^2 + 2z^2 = 1.$$

Introduce $u = x$ and $v = \sqrt{2}\,z$. Then $u^2 + v^2 = 1$ and

$$f = 4u - 2\sqrt{2}\,v.$$

Its maximum and minimum on the unit circle are

$$\pm\sqrt{4^2 + (2\sqrt{2})^2} = \pm 2\sqrt{6}.$$

So

$$f_{\max} = 2\sqrt{6}, \qquad f_{\min} = -2\sqrt{6}.$$

The maximizing point on the (u,v)-circle is

$$(u,v) = \frac{(4, -2\sqrt{2})}{2\sqrt{6}} = \left(\frac{\sqrt{6}}{3}, -\frac{1}{\sqrt{3}}\right).$$

Hence

$$x = \frac{\sqrt{6}}{3}, \qquad z = \frac{v}{\sqrt{2}} = -\frac{1}{\sqrt{6}}, \qquad y = z - x = -\frac{\sqrt{6}}{2}.$$

Thus

$$f_{\max} = 2\sqrt{6} \text{ at } \left(\frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{2}, -\frac{1}{\sqrt{6}}\right).$$

By symmetry,

$$f_{\min} = -2\sqrt{6} \text{ at } \left(-\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{2}, \frac{1}{\sqrt{6}}\right).$$

(iii) $f(x,y,z) = yz + xy$ subject to $xy = 1$ and $y^2 + z^2 = 1$.

Solution. From $xy = 1$ we have $x = 1/y$ (so $y \neq 0$). Then

$$f = yz + xy = yz + 1.$$

Thus we only need to maximize/minimize yz subject to

$$y^2 + z^2 = 1.$$

Since

$$2|yz| \leq y^2 + z^2 = 1,$$

we get

$$-\frac{1}{2} \le yz \le \frac{1}{2}.$$

Therefore

$$f_{\max} = 1 + \frac{1}{2} = \frac{3}{2}, \qquad f_{\min} = 1 - \frac{1}{2} = \frac{1}{2}.$$

The maximum $yz = \frac{1}{2}$ occurs at

$$(y, z) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right),$$

and then $x = 1/y = \pm\sqrt{2}$. Hence

$$f_{\max} = \frac{3}{2} \text{ at } \left(\sqrt{2}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(-\sqrt{2}, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right).$$

The minimum $yz = -\frac{1}{2}$ occurs at

$$(y, z) = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right),$$

so

$$f_{\min} = \frac{1}{2} \text{ at } \left(\sqrt{2}, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(-\sqrt{2}, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

(iv) $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x - y = 1$ and $y^2 - z^2 = 1$.

Solution. From the constraints,

$$x = y + 1, \qquad z^2 = y^2 - 1.$$

Hence

$$f = (y + 1)^2 + y^2 + (y^2 - 1) = 3y^2 + 2y.$$

The constraint $z^2 = y^2 - 1 \ge 0$ implies $|y| \ge 1$. So we minimize $3y^2 + 2y$ on the set $(-\infty, -1] \cup [1, \infty)$.

Since $3y^2 + 2y$ is an upward-opening parabola, and its vertex $y = -1/3$ is not feasible, the minimum occurs at the nearest feasible endpoint $y = -1$. Then

$$x = 0, \qquad z = 0, \qquad f = 1.$$

As $|y| \to \infty$, we have $f = 3y^2 + 2y \to \infty$, so there is no maximum.

Therefore

$$f_{\min} = 1 \text{ at } (0, -1, 0),$$

and there is no maximum.

Problem 2

Find the absolute maximum and minimum values of the following function in the specified region.

(i) $f(x, y) = 4xy^2 - x^2y^2 - xy^3$ on the closed triangle with vertices $(0, 0)$, $(0, 6)$, $(6, 0)$.

Solution. The region is

$$R = \{(x, y) : x \geq 0,\ y \geq 0,\ x + y \leq 6\}.$$

Write

$$f(x, y) = xy^2(4 - x - y).$$

Since R is compact and f is continuous, absolute extrema exist.

Interior critical points.

$$f_x = y^2(4 - 2x - y), \qquad f_y = xy(8 - 2x - 3y).$$

In the interior, $x > 0$ and $y > 0$, so

$$4 - 2x - y = 0, \qquad 8 - 2x - 3y = 0.$$

Subtracting gives $2y = 4$, so $y = 2$, and then $x = 1$. Thus the only interior critical point is $(1, 2)$, where

$$f(1, 2) = 4.$$

Boundary. On $x = 0$ or $y = 0$, we have $f = 0$.

On the slanted edge $x + y = 6$, put $y = 6 - x$:

$$f(x, 6 - x) = x(6 - x)^2(4 - 6) = -2x(6 - x)^2, \qquad 0 \leq x \leq 6.$$

Let

$$g(x) = -2x(6 - x)^2.$$

Then

$$g'(x) = -2(6 - x)(6 - 3x).$$

So critical points are $x = 2$ and $x = 6$. Evaluating,

$$g(0) = 0, \qquad g(2) = -64, \qquad g(6) = 0.$$

Hence the boundary minimum is -64 at $(2, 4)$.

Comparing all candidates:

$$f_{\max} = 4 \text{ at } (1, 2), \qquad f_{\min} = -64 \text{ at } (2, 4).$$

(ii) $f(x, y) = e^{-x^2-y^2}(x^2 + 2y^2)$ on the disk $x^2 + y^2 \leq 1$.

Solution. Write $x = r\cos\theta$, $y = r\sin\theta$. Then

$$f(r, \theta) = e^{-r^2} r^2(\cos^2\theta + 2\sin^2\theta).$$

For fixed θ, the angular factor is positive and

$$g(r) = r^2 e^{-r^2}, \qquad 0 \leq r \leq 1.$$

Now

$$g'(r) = 2re^{-r^2}(1 - r^2) \geq 0 \quad \text{for } 0 \leq r \leq 1.$$

So for each θ, the maximum occurs at $r = 1$ and the minimum at $r = 0$.

At $r = 0$,

$$f(0, 0) = 0.$$

At $r = 1$,

$$f = e^{-1}(\cos^2\theta + 2\sin^2\theta) = e^{-1}(1 + \sin^2\theta).$$

This is largest when $\sin^2\theta = 1$, i.e. at $(0, \pm 1)$, and smallest on the boundary when $\sin^2\theta = 0$, i.e. at $(\pm 1, 0)$.

Therefore

$$f_{\text{min}} = 0 \text{ at } (0, 0),$$

$$f_{\text{max}} = \frac{2}{e} \text{ at } (0, 1) \text{ and } (0, -1).$$

Problem 3

Evaluate the double integral by first identifying it as the volume of a solid.

(i) $\iint_R 3\, dA$, where $R = \{(x, y) : -2 \leq x \leq 2,\ 1 \leq y \leq 6\}$.

Solution. This is the volume of the solid of constant height 3 over the rectangle R. The area of the base is

$$\text{Area}(R) = (2 - (-2))(6 - 1) = 4 \cdot 5 = 20.$$

Hence

$$\iint_R 3\, dA = 3 \cdot 20 = 60.$$

(ii) $\iint_R 5x\, dA$, where $R = \{(x, y) : 0 \leq x \leq 5,\ 0 \leq y \leq 3\}$.

Solution. This is the volume under the plane $z = 5x$ over the given rectangle:

$$\iint_R 5x\, dA = \int_0^5 \int_0^3 5x\, dy\, dx = \int_0^5 15x\, dx = 15 \cdot \frac{25}{2} = \frac{375}{2}.$$

Problem 4

Evaluate the following iterated integrals.

(i) $\int_0^2 \int_0^4 y^2 e^{2x}\, dy\, dx$.

$$\int_0^2 e^{2x} \left(\int_0^4 y^2\, dy \right) dx = \int_0^2 e^{2x} \cdot \frac{64}{3}\, dx = \frac{64}{3} \cdot \frac{e^4 - 1}{2} = \frac{32}{3}(e^4 - 1).$$

(ii) $\displaystyle\int_{-3}^{3}\int_{0}^{\pi/2}(y+y^2\cos x)\,dx\,dy.$

$$\int_{-3}^{3}\left(\frac{\pi}{2}y+y^2\int_0^{\pi/2}\cos x\,dx\right)dy=\int_{-3}^{3}\left(\frac{\pi}{2}y+y^2\right)dy.$$

The first term is odd and integrates to 0:

$$\int_{-3}^{3}\frac{\pi}{2}y\,dy=0.$$

Thus

$$\int_{-3}^{3}y^2\,dy=2\int_0^3 y^2\,dy=2\cdot 9=18.$$

So the value is

$$18.$$

(iii) $\displaystyle\int_1^3\int_1^5(\ln y)\,xy\,dy\,dx.$

$$\left(\int_1^3 x\,dx\right)\left(\int_1^5 y\ln y\,dy\right)=4\left[\frac{y^2}{2}\ln y-\frac{y^2}{4}\right]_1^5.$$

Hence

$$4\left(\frac{25}{2}\ln 5-\frac{25}{4}+\frac{1}{4}\right)=50\ln 5-24.$$

(iv) $\displaystyle\int_0^1\int_0^1 v(u+v^2)^4\,du\,dv.$

$$\int_0^1 v\left[\frac{(u+v^2)^5}{5}\right]_{u=0}^{u=1}dv=\frac{1}{5}\int_0^1 v\left((1+v^2)^5-v^{10}\right)dv.$$

Now

$$\int_0^1 v(1+v^2)^5\,dv=\frac{1}{2}\int_1^2 t^5\,dt=\frac{1}{12}(64-1)=\frac{63}{12},$$

and

$$\int_0^1 v^{11}\,dv=\frac{1}{12}.$$

Therefore

$$\frac{1}{5}\left(\frac{63}{12}-\frac{1}{12}\right)=\frac{1}{5}\cdot\frac{62}{12}=\frac{21}{20}.$$

(v) $\displaystyle\int_0^1\int_0^1 xy\sqrt{x^2+y^2}\,dy\,dx.$

Integrate with respect to y first:

$$\int_0^1 x\left(\int_0^1 y\sqrt{x^2+y^2}\,dy\right)dx.$$

Let $t=x^2+y^2$, so $dt=2y\,dy$:

$$\int y\sqrt{x^2+y^2}\,dy=\frac{1}{2}\int t^{1/2}\,dt=\frac{1}{3}t^{3/2}.$$

Hence

$$\int_0^1 x \cdot \frac{1}{3}\left[(x^2+y^2)^{3/2}\right]_{y=0}^{y=1} dx = \frac{1}{3}\int_0^1 x\Big((x^2+1)^{3/2} - x^3\Big)\, dx.$$

Now

$$\int_0^1 x(x^2+1)^{3/2}\, dx = \frac{1}{2}\int_1^2 t^{3/2}\, dt = \frac{1}{5}\big(2^{5/2}-1\big) = \frac{4\sqrt{2}-1}{5},$$

and

$$\int_0^1 x^4\, dx = \frac{1}{5}.$$

Therefore

$$\frac{1}{3}\left(\frac{4\sqrt{2}-1}{5} - \frac{1}{5}\right) = \frac{2(2\sqrt{2}-1)}{15}.$$

(vi) $\displaystyle\int_0^2 \int_0^\pi r\sin^2\theta\, d\theta\, dr.$

$$\left(\int_0^2 r\, dr\right)\left(\int_0^\pi \sin^2\theta\, d\theta\right) = 2\cdot\frac{\pi}{2} = \pi.$$

(vii) $\displaystyle\int_0^1 \int_0^1 \sqrt{s+t}\, ds\, dt.$

Integrating in s first,

$$\int_0^1 \left[\frac{2}{3}(s+t)^{3/2}\right]_{s=0}^{s=1} dt = \frac{2}{3}\int_0^1 \big((1+t)^{3/2} - t^{3/2}\big)\, dt.$$

So

$$\frac{2}{3}\left(\frac{2}{5}(2^{5/2}-1) - \frac{2}{5}\right) = \frac{8(2\sqrt{2}-1)}{15}.$$

(viii) $\displaystyle\int_1^2 \int_0^{2z} \int_0^{\ln x} xe^y\, dy\, dx\, dz.$

First integrate in y:

$$\int_0^{\ln x} xe^y\, dy = x(e^{\ln x}-1) = x(x-1) = x^2 - x.$$

So

$$\int_1^2 \int_0^{2z} (x^2-x)\, dx\, dz = \int_1^2 \left[\frac{x^3}{3} - \frac{x^2}{2}\right]_0^{2z} dz = \int_1^2 \left(\frac{8z^3}{3} - 2z^2\right) dz.$$

This gives

$$\left[\frac{2z^4}{3} - \frac{2z^3}{3}\right]_1^2 = \frac{16}{3}.$$

(ix) $\displaystyle\int_0^{\pi/2} \int_0^y \int_0^x \cos(x+y+z)\, dz\, dx\, dy.$

First,

$$\int_0^x \cos(x+y+z)\, dz = \sin(x+y+z)\Big|_0^x = \sin(2x+y) - \sin(x+y).$$

So the integral becomes

$$\int_0^{\pi/2} \int_0^y \big(\sin(2x+y) - \sin(x+y)\big)\, dx\, dy.$$

Integrating in x,

$$\int_0^y \sin(2x+y)\, dx = -\frac{1}{2}\cos(2x+y)\Big|_0^y = -\frac{1}{2}\cos 3y + \frac{1}{2}\cos y,$$

$$\int_0^y \sin(x+y)\, dx = -\cos(x+y)\Big|_0^y = -\cos 2y + \cos y.$$

Hence the inner integral is

$$-\frac{1}{2}\cos 3y + \cos 2y - \frac{1}{2}\cos y.$$

Now integrate from 0 to $\pi/2$:

$$\left[-\frac{1}{6}\sin 3y + \frac{1}{2}\sin 2y - \frac{1}{2}\sin y\right]_0^{\pi/2} = \frac{1}{6} - \frac{1}{2} = -\frac{1}{3}.$$

Therefore the value is

$$-\frac{1}{3}.$$

(x) $\displaystyle\int_0^{\sqrt{\pi}} \int_0^x \int_0^{xz} x^2 \sin y\, dy\, dz\, dx.$

First,

$$\int_0^{xz} x^2 \sin y\, dy = x^2(1 - \cos(xz)).$$

Then

$$\int_0^x x^2(1-\cos(xz))\, dz = x^2\left[z - \frac{\sin(xz)}{x}\right]_0^x = x^3 - x\sin(x^2).$$

Therefore

$$\int_0^{\sqrt{\pi}} \big(x^3 - x\sin(x^2)\big)\, dx = \left[\frac{x^4}{4}\right]_0^{\sqrt{\pi}} - \int_0^{\sqrt{\pi}} x\sin(x^2)\, dx.$$

For the second term, let $u = x^2$, $du = 2x\, dx$:

$$\int_0^{\sqrt{\pi}} x\sin(x^2)\, dx = \frac{1}{2}\int_0^{\pi} \sin u\, du = 1.$$

Hence the value is

$$\frac{\pi^2}{4} - 1.$$

Problem 5

Calculate the double integral of the following functions in the specified regions.

(i) $f(x,y) = \sin(x-y)$ on $R = \{(x,y) : 0 \le x \le \pi/2,\ 0 \le y \le \pi/2\}$.

Solution. Since the square is symmetric under $(x,y) \leftrightarrow (y,x)$ and

$$\sin(y-x) = -\sin(x-y),$$

the integral is zero:

$$\iint_R \sin(x-y)\,dA = 0.$$

(ii) $f(x,y) = y + xy^{-2}$ on $R = \{(x,y) : 0 \le x \le 2,\ 1 \le y \le 2\}$.

Solution.

$$\int_0^2 \int_1^2 \left(y + \frac{x}{y^2}\right) dy\,dx = \int_0^2 \left[\frac{y^2}{2} - \frac{x}{y}\right]_1^2 dx = \int_0^2 \left(\frac{3}{2} + \frac{x}{2}\right) dx = 4.$$

(iii) $f(x,y) = x^2 + 2y$, where R is bounded by $y = 0$, $y = x^2$, $x = 1$.

Solution. The region is

$$0 \le x \le 1, \qquad 0 \le y \le x^2.$$

Thus

$$\iint_R (x^2 + 2y)\,dA = \int_0^1 \int_0^{x^2} (x^2 + 2y)\,dy\,dx = \int_0^1 \left(x^4 + x^4\right) dx = 2\int_0^1 x^4\,dx = \frac{2}{5}.$$

(iv) $f(x,y) = y^2$, where R is the triangle with vertices $(0,1)$, $(1,2)$, $(4,1)$.

Solution. The two nonhorizontal sides are

$$y = x + 1 \quad (0 \le x \le 1), \qquad y = \frac{7-x}{3} \quad (1 \le x \le 4).$$

Hence

$$\iint_R y^2\,dA = \int_0^1 \int_1^{x+1} y^2\,dy\,dx + \int_1^4 \int_1^{(7-x)/3} y^2\,dy\,dx.$$

Compute:

$$\int_0^1 \int_1^{x+1} y^2\,dy\,dx = \frac{1}{3}\int_0^1 \left((x+1)^3 - 1\right) dx = \frac{11}{12},$$

$$\int_1^4 \int_1^{(7-x)/3} y^2\,dy\,dx = \frac{1}{3}\int_1^4 \left(\left(\frac{7-x}{3}\right)^3 - 1\right) dx = \frac{11}{4}.$$

Therefore

$$\iint_R y^2\,dA = \frac{11}{12} + \frac{11}{4} = \frac{11}{3}.$$

(v) $f(x,y) = xy^2$, where R is enclosed by $x = 0$ and $x = \sqrt{1-y^2}$.

Solution. This is the right half of the unit disk:

$$-1 \le y \le 1, \qquad 0 \le x \le \sqrt{1-y^2}.$$

Thus

$$\iint_R xy^2\,dA = \int_{-1}^{1}\int_0^{\sqrt{1-y^2}} xy^2\,dx\,dy = \int_{-1}^{1} y^2\left[\frac{x^2}{2}\right]_0^{\sqrt{1-y^2}} dy.$$

Hence

$$= \frac{1}{2}\int_{-1}^{1}(y^2-y^4)\,dy = \int_0^1 (y^2-y^4)\,dy = \frac{1}{3}-\frac{1}{5} = \frac{2}{15}.$$

(vi) $f(x,y) = \sin(x^2+y^2)$, where R is the region in the first quadrant between the circles $x^2+y^2 = a^2$ and $x^2+y^2 = b^2$, with $0 < a < b$.

Solution. Use polar coordinates:

$$0 \le \theta \le \frac{\pi}{2}, \qquad a \le r \le b.$$

Then

$$\iint_R \sin(x^2+y^2)\,dA = \int_0^{\pi/2}\int_a^b \sin(r^2)\,r\,dr\,d\theta.$$

Let $u = r^2$, $du = 2r\,dr$:

$$\int_a^b \sin(r^2)\,r\,dr = \frac{1}{2}\int_{a^2}^{b^2}\sin u\,du = \frac{1}{2}\big(\cos a^2 - \cos b^2\big).$$

Therefore

$$\iint_R \sin(x^2+y^2)\,dA = \frac{\pi}{4}\big(\cos a^2 - \cos b^2\big).$$

(vii) $f(x,y) = e^{-x^2-y^2}$, where R is bounded by the semicircle $x = \sqrt{4-y^2}$ and the y-axis.

Solution. This is the right semicircle of radius 2:

$$0 \le r \le 2, \qquad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}.$$

Thus

$$\iint_R e^{-x^2-y^2}\,dA = \int_{-\pi/2}^{\pi/2}\int_0^2 e^{-r^2}\,r\,dr\,d\theta.$$

Now

$$\int_0^2 e^{-r^2}\,r\,dr = \frac{1}{2}\int_0^4 e^{-u}\,du = \frac{1}{2}(1-e^{-4}).$$

Multiplying by the angular length π gives

$$\iint_R e^{-x^2-y^2}\,dA = \frac{\pi}{2}(1-e^{-4}).$$

(viii) $f(x,y) = \arctan\left(\frac{y}{x}\right)$ on $R = \{(x,y) : 1 \le x^2 + y^2 \le 4,\ 0 \le y \le x\}$.

Solution. In polar coordinates, the region is

$$1 \le r \le 2, \qquad 0 \le \theta \le \frac{\pi}{4},$$

and

$$\arctan\left(\frac{y}{x}\right) = \theta.$$

Therefore

$$\iint_R \arctan\left(\frac{y}{x}\right)\, dA = \int_0^{\pi/4} \int_1^2 \theta r\, dr\, d\theta = \left(\int_1^2 r\, dr\right)\left(\int_0^{\pi/4} \theta\, d\theta\right).$$

Hence

$$= \frac{3}{2} \cdot \frac{\pi^2}{32} = \frac{3\pi^2}{64}.$$

Problem 6

Solve the following volume problems.

(i) Volume bounded by the circular paraboloid $z = 1 - x^2 - y^2$ and the xy-plane.

Solution. The paraboloid meets the plane $z = 0$ when

$$x^2 + y^2 = 1.$$

So in polar coordinates,

$$0 \le r \le 1, \qquad 0 \le \theta \le 2\pi.$$

The volume is

$$V = \int_0^{2\pi} \int_0^1 (1 - r^2) r\, dr\, d\theta = 2\pi \left[\frac{r^2}{2} - \frac{r^4}{4}\right]_0^1 = 2\pi\left(\frac{1}{4}\right) = \frac{\pi}{2}.$$

(ii) Volume under the cone $z = \sqrt{x^2 + y^2}$ and above the disk $x^2 + y^2 \le 4$.

Solution. In polar coordinates, $z = r$ and $0 \le r \le 2$:

$$V = \int_0^{2\pi} \int_0^2 r \cdot r\, dr\, d\theta = 2\pi \int_0^2 r^2\, dr = 2\pi \cdot \frac{8}{3} = \frac{16\pi}{3}.$$

(iii) Volume below the paraboloid $z = 18 - 2x^2 - 2y^2$ and above the xy-plane.

Solution. The paraboloid meets $z = 0$ when

$$18 - 2r^2 = 0 \quad \Longrightarrow \quad r = 3.$$

Hence

$$V = \int_0^{2\pi} \int_0^3 (18 - 2r^2) r\, dr\, d\theta = 2\pi \left[9r^2 - \frac{r^4}{2}\right]_0^3 = 2\pi\left(81 - \frac{81}{2}\right) = 81\pi.$$

(iv) Volume enclosed by the hyperboloid $-x^2 - y^2 + z^2 = 1$ and the plane $z = 2$.

Solution. Rewrite the surface as

$$z^2 - r^2 = 1 \quad \Longrightarrow \quad r^2 = z^2 - 1.$$

For $1 \le z \le 2$, the cross-section at height z is a disk of area

$$\pi(z^2 - 1).$$

Thus

$$V = \int_1^2 \pi(z^2 - 1)\, dz = \pi \left[\frac{z^3}{3} - z \right]_1^2 = \frac{4\pi}{3}.$$

(v) Volume inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the cylinder $x^2 + y^2 = 4$.

Solution. In cylindrical coordinates, the region is

$$2 \le r \le 4, \qquad 0 \le \theta \le 2\pi, \qquad -\sqrt{16 - r^2} \le z \le \sqrt{16 - r^2}.$$

Hence

$$V = \int_0^{2\pi} \int_2^4 \int_{-\sqrt{16-r^2}}^{\sqrt{16-r^2}} r\, dz\, dr\, d\theta = 4\pi \int_2^4 r\sqrt{16 - r^2}\, dr.$$

Let $u = 16 - r^2$, so $du = -2r\, dr$:

$$V = 4\pi \cdot \frac{1}{2} \int_0^{12} u^{1/2}\, du = 2\pi \cdot \frac{2}{3} u^{3/2} \Big|_0^{12} = \frac{4\pi}{3} \cdot 12^{3/2}.$$

Since $12^{3/2} = 24\sqrt{3}$,

$$V = 32\pi\sqrt{3}.$$

Problem 7

Evaluate the triple integral of the following functions in the specified solids.

(i) $f(x, y, z) = y, \quad V = \{(x, y, z) : 0 \le x \le 3,\ 0 \le y \le x,\ x - y \le z \le x + y\}$.

Solution.

$$\iiint_V y\, dV = \int_0^3 \int_0^x \int_{x-y}^{x+y} y\, dz\, dy\, dx.$$

Since the z-length is

$$(x + y) - (x - y) = 2y,$$

we get

$$\iiint_V y\, dV = \int_0^3 \int_0^x 2y^2\, dy\, dx = 2 \int_0^3 \frac{x^3}{3}\, dx = \frac{2}{3} \cdot \frac{81}{4} = \frac{27}{2}.$$

(ii) $f(x, y, z) = \dfrac{z}{x^2 + y^2}, \quad V = \{1 \le y \le 4,\ y \le z \le 4,\ 0 \le x \le z\}$.

Solution. We write

$$I = \int_1^4 \int_y^4 \int_0^z \frac{z}{x^2 + y^2}\, dx\, dz\, dy.$$

Integrating in x first gives

$$\int_0^z \frac{z}{x^2+y^2}\,dx = \frac{z}{y}\arctan\left(\frac{z}{y}\right).$$

Hence

$$I = \int_1^4 \int_y^4 \frac{z}{y}\arctan\left(\frac{z}{y}\right)\,dz\,dy.$$

Now set $t = z/y$. Then $z = yt$, $dz = y\,dt$, and for fixed y, t runs from 1 to $4/y$. So

$$I = \int_1^4 y\left(\int_1^{4/y} t\arctan t\,dt\right)dy.$$

Swapping the order in the (y,t)-plane gives

$$I = \frac{1}{2}\int_1^4 \left(\frac{16}{t} - t\right)\arctan t\,dt.$$

The term $\int t\arctan t\,dt$ is elementary, while $\int \frac{\arctan t}{t}\,dt$ is the inverse tangent integral $\mathrm{Ti}_2(t) = \int_0^t \frac{\arctan u}{u}\,du$. Therefore

$$I = 8\big(\mathrm{Ti}_2(4) - \mathrm{Ti}_2(1)\big) - \frac{1}{4}\left(17\arctan 4 - \frac{\pi}{2} - 3\right).$$

Numerically,

$$I \approx 7.58735.$$

(iii) $f(x,y,z) = \sin y$, where V is the solid below the plane $z = x$ and above the triangular region with vertices $(0,0,0)$, $(\pi,0,0)$, $(0,\pi,0)$.

Solution. The base in the xy-plane is

$$D = \{(x,y) : x \geq 0,\ y \geq 0,\ x + y \leq \pi\},$$

and z runs from 0 to x. Hence

$$\iiint_V \sin y\,dV = \int_0^\pi \int_0^{\pi-y} \int_0^x \sin y\,dz\,dx\,dy = \int_0^\pi \int_0^{\pi-y} x\sin y\,dx\,dy.$$

Thus

$$= \frac{1}{2}\int_0^\pi (\pi - y)^2 \sin y\,dy.$$

Expanding,

$$\frac{1}{2}\int_0^\pi (\pi^2 - 2\pi y + y^2)\sin y\,dy.$$

Using

$$\int_0^\pi \sin y\,dy = 2, \qquad \int_0^\pi y\sin y\,dy = \pi, \qquad \int_0^\pi y^2\sin y\,dy = \pi^2 - 4,$$

we get

$$\iiint_V \sin y\,dV = \frac{1}{2}(\pi^2 - 4).$$

(iv) $f(x,y,z) = xy$, where V is bounded by the parabolic cylinders $y = x^2$ and $x = y^2$, and the planes $z = 0$ and $z = x + y$.

Solution. In the first quadrant, the curves $y = x^2$ and $x = y^2$ meet at $(0,0)$ and $(1,1)$. For $0 \le x \le 1$, we have

$$x^2 \le y \le \sqrt{x}.$$

Since $0 \le z \le x + y$, the integral is

$$\iiint_V xy\,dV = \int_0^1 \int_{x^2}^{\sqrt{x}} \int_0^{x+y} xy\,dz\,dy\,dx = \int_0^1 \int_{x^2}^{\sqrt{x}} xy(x+y)\,dy\,dx.$$

That is,

$$\int_0^1 \int_{x^2}^{\sqrt{x}} (x^2y + xy^2)\,dy\,dx.$$

Integrating in y:

$$\int_0^1 \left[\frac{x^2y^2}{2} + \frac{xy^3}{3}\right]_{y=x^2}^{y=\sqrt{x}} dx = \int_0^1 \left(\frac{x^3}{2} + \frac{x^{5/2}}{3} - \frac{x^6}{2} - \frac{x^7}{3}\right) dx.$$

So

$$\iiint_V xy\,dV = \frac{1}{8} + \frac{2}{21} - \frac{1}{14} - \frac{1}{24} = \frac{3}{28}.$$

(v) $f(x,y,z) = x^2$, where V is the tetrahedron with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, $(0,0,1)$.

Solution. The tetrahedron is

$$V = \{(x,y,z) : x \ge 0,\ y \ge 0,\ z \ge 0,\ x + y + z \le 1\}.$$

For fixed x, the (y,z)-section is a right triangle of area

$$\frac{(1-x)^2}{2}.$$

Therefore

$$\iiint_V x^2\,dV = \int_0^1 x^2 \cdot \frac{(1-x)^2}{2}\,dx = \frac{1}{2}\int_0^1 (x^2 - 2x^3 + x^4)\,dx.$$

Hence

$$\iiint_V x^2\,dV = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5}\right) = \frac{1}{60}.$$

Problem 8

For each repeated integral, describe the region and change the order of integration.

(i) $\displaystyle\int_0^1 \left[\int_0^y f(x,y)\,dx\right] dy.$

Region.

$$R = \{(x,y) : 0 \le x \le y \le 1\}.$$

Reversing the order, for fixed x, y runs from x to 1. Therefore

$$\int_0^1 \int_x^1 f(x,y)\,dy\,dx.$$

(ii) $\displaystyle\int_0^2 \left[\int_{2y}^{y^2} f(x,y)\,dx\right] dy.$

Region. The geometric region is bounded by

$$x = 2y, \qquad x = y^2, \qquad 0 \le y \le 2.$$

Since $y^2 \le 2y$ on $[0,2]$, the original integral has reversed x-limits. The corresponding geometric region is

$$R = \{(x,y) : 0 \le x \le 4,\ x/2 \le y \le \sqrt{x}\}.$$

Hence the same signed integral can be written as

$$\int_0^4 \int_{\sqrt{x}}^{x/2} f(x,y)\,dy\,dx,$$

or equivalently

$$-\int_0^4 \int_{x/2}^{\sqrt{x}} f(x,y)\,dy\,dx.$$

(iii) Interpreting the third integral consistently as

$$\int_1^e \left[\int_0^{\ln x} f(x,y)\,dy\right] dx.$$

Region.

$$R = \{(x,y) : 1 \le x \le e,\ 0 \le y \le \ln x\}.$$

For fixed y, this becomes

$$e^y \le x \le e, \qquad 0 \le y \le 1.$$

So the reversed order is

$$\int_0^1 \int_{e^y}^{e} f(x,y)\,dx\,dy.$$

Problem 9

Write three other iterated integrals equal to the given one.

(i) Given

$$\int_0^1 \int_y^1 \int_0^y f(x,y,z)\,dx\,dz\,dy.$$

Region.

$$0 \le x \le y \le z \le 1.$$

Three other equivalent iterated integrals are:

$$\int_0^1 \int_0^z \int_x^z f(x,y,z)\,dy\,dx\,dz,$$

$$\int_0^1 \int_x^1 \int_x^z f(x,y,z)\,dy\,dz\,dx,$$

$$\int_0^1 \int_x^1 \int_y^1 f(x,y,z)\,dz\,dy\,dx.$$

(ii) Given

$$\int_0^1 \int_0^{1-x^2} \int_0^{1-x} f(x,y,z)\,dy\,dz\,dx.$$

Region.

$$0 \le x \le 1, \qquad 0 \le y \le 1-x, \qquad 0 \le z \le 1-x^2.$$

Three other equivalent iterated integrals are:

$$\int_0^1 \int_0^{1-x} \int_0^{1-x^2} f(x,y,z)\,dz\,dy\,dx,$$

$$\int_0^1 \int_0^{1-y} \int_0^{1-x^2} f(x,y,z)\,dz\,dx\,dy,$$

$$\int_0^1 \int_0^{\sqrt{1-z}} \int_0^{1-x} f(x,y,z)\,dy\,dx\,dz.$$

(iii) Given

$$\int_0^1 \int_{\sqrt{x}}^1 \int_0^{1-y} f(x,y,z)\,dz\,dy\,dx.$$

Region. Since $\sqrt{x} \le y$ is equivalent to $x \le y^2$, the region is

$$0 \le y \le 1, \qquad 0 \le x \le y^2, \qquad 0 \le z \le 1-y.$$

Three other equivalent iterated integrals are:

$$\int_0^1 \int_0^{1-y} \int_0^{y^2} f(x,y,z)\,dx\,dz\,dy,$$

$$\int_0^1 \int_0^{1-z} \int_0^{y^2} f(x,y,z)\,dx\,dy\,dz,$$

$$\int_0^1 \int_0^{(1-z)^2} \int_{\sqrt{x}}^{1-z} f(x,y,z)\,dy\,dx\,dz.$$